JOSEPH ALAN B. VERGARA

Embedded Systems | Edge AI | Firmware | Applied Computing | Research
Iligan City, Philippines • +63 947-589-2995 • josephalan.vergara@g.msuiit.edu.ph
hpesojnalab.aragrev@gmail.com
Portfolio • LinkedIn • GitHub

Academic Profile

Joseph Alan B. Vergara is a fourth-year BS Computer Applications student majoring in Embedded Systems at Mindanao State University – Iligan Institute of Technology. His work focuses on microcontroller firmware, real-time operating systems, edge AI, computer vision, and sensor telemetry. He contributes to PhD, master's, and undergraduate research collaborations, developing embedded systems, mobile applications, and web platforms that connect hardware with practical applications.

Education

- **BS Computer Applications, Major in Embedded Systems (CGPA: 1.96269) —** Mindanao State University – Iligan Institute of Technology | August 2023 – Present; fourth year.
- **Senior High School, Accountancy, Business & Management (Academic Honors) —** MSU-IIT Integrated Developmental School | 2021 – 2023.

Research & Development Experience

- **Co-Author & Full-Stack / Embedded Developer —** PhD dissertation research collaboration | 2024 – Present. Developed a React, Node.js, and PostgreSQL learning platform for microcontroller telemetry and automated laboratory scoring; benchmarked cloud and ESP32-P4 edge-hosted architectures. In progress. GitHub · Live demo
- **Co-Author & Mobile / Edge AI Developer —** Master's and undergraduate research collaboration | 2024 – Present. Deployed an optimized YOLOv8n model for on-device microplastic detection; implemented OpenCV preprocessing, camera lifecycles, particle visualization, and Android release builds. In progress. GitHub · Documentation

Projects

- **IoT Human Activity Recognition & Edge Telemetry (BCA182 Lab 3) —** Built an STM32/RT-Spark activity-recognition pipeline with IMU feature extraction, MQTT ingestion, and a telemetry dashboard. GitHub
- **ARUGA Fall Detection & Inactivity Monitoring —** Developed an edge computer-vision system for fall detection and inactivity monitoring. GitHub
- **Personal MP3 Player & Tone Sequencer —** Implemented a FreeRTOS audio sequencer with LCD rendering, button decoding, ADC volume control, and 18 unit tests. GitHub
- **O.I.N.K. Swine Thermal Fever Surveillance —** Built ESP32 thermal streaming with an MLX90640 sensor, OpenCV processing, and a Flutter monitoring dashboard. Live demo
- **Agap AI Emergency Triage Platform —** Developed an emergency-response platform that converts voice reports into structured, triage-ready incident data. Live demo
- **Multisensor Room Monitor —** Implemented five FreeRTOS tasks for sensor acquisition, OLED output, encoder navigation, logging, and power management. GitHub · Write-up
- **Microcontroller-Based Automated Feedback System —** Developed a research learning platform for microcontroller telemetry and automated laboratory scoring, including ESP32-P4 edge-hosting benchmarks. GitHub · Live demo
- **Mobile Microplastic Detection App —** Implemented on-device YOLOv8n detection, OpenCV preprocessing, camera visualization, and mobile release builds for microplastic research. GitHub · Documentation
- **Barangay Connect Web Application —** Built a React and Supabase community portal for incident reports, document requests, polls, and notifications. GitHub · Live demo

Projects (continued)

- **Snake OS: Embedded Interactive HMI Game —** Developed a Renesas RA6M3 arcade game using RT-Thread and LVGL, with AI navigation and persistent high scores. GitHub · Write-up
- **Sustainability in the Cup Web —** Built a sustainability research site with green-claim evaluation, survey instruments, and data visualization. Live demo
- **Edge AI Glasses Object Detection —** Trained and quantized a wearable object-detection model using Roboflow datasets and Edge Impulse. Project
- **Industrial Oven Simulation & Control —** Engineered CODESYS ladder logic and an HMI with heating control, safety interlocks, and threshold alarms. GitHub · Documentation
- **IoT Vermicomposting Monitor —** Implemented ESP32 soil-moisture monitoring, closed-loop irrigation, and environmental telemetry over MQTT. Documentation
- **Line Following Robot —** Programmed AVR Assembly firmware and dual PWM timers for autonomous line tracking using a five-channel IR array. Documentation
- **K-Bin: Automated Smart Waste Sorting —** Built a Qt C++ interface and sensor-fusion system for three waste streams, with SQLite maintenance logging. Documentation
- **Adjustable DC Power Supply —** Designed a regulated variable-output supply up to 12 V with short-circuit protection for embedded prototyping. Portfolio

Skills & Interests

- **Programming —** C, C++, Python, JavaScript, TypeScript, Java, HTML, CSS, Bash, Assembly, and PLC ladder logic.
- **Embedded Systems & Hardware —** STM32, ESP32/ESP32-P4, AVR, Renesas RA6M3, 8051, FreeRTOS, RT-Thread, PCB layout, simulation, soldering, test equipment, and computer hardware repair.
- **AI, Software & Tools —** YOLOv8, OpenCV, MediaPipe, Edge Impulse, TensorFlow Lite, MATLAB, React, Next.js, Node.js, Flutter, Supabase/PostgreSQL, MongoDB, Git, Arduino IDE, Android SDK, SquareLine Studio, and Vivado.
- **Systems & Infrastructure —** Linux, shell configuration, DNS, SSL/TLS, Vercel, Railway, Hostinger, Cisco Packet Tracer, MQTT, I2C, SPI, UART, and CAN.
- **Professional Skills & Languages —** Cross-functional collaboration, time management, independent learning, and adaptability. English, Filipino, and Cebuano.

Leadership

- **Undersecretary, Sports & Development Affairs —** College of Computer Studies Executive Council, MSU-IIT | August 2025 – May 2026.
- **Team Captain, Mobile Legends: Bang Bang —** Titans Esports | 2021 – 2023.

Awards & Recognitions

- **IEEE Sumpai Hackathon 2026 —** 3rd Place and Top 5 Finalist; developed Agap AI | September 14, 2026.
- **TechShowcase 2025 —** 2nd Best Innovation and 3rd Best Pitch; developed Barangay Connect | December 17, 2025.

Certifications

- **Startup Basics 101 —** iDEYA: Center of Innovation and Technopreneurship, MSU-IIT | September 24, 2026.
- **AI Career Readiness Training —** MSU-IIT | August 20, 2026.
- **Quantum Circuits as Pictures: An Introduction to ZX Calculus —** Webinar | March 20, 2026.
- **Permaculture: “From Code to Crops” —** Webinar | December 22, 2025.
- **my.ComApps Technology Symposium: RT-Thread Based Technology —** MSU-IIT | November 24–25, 2025.
- **Seize Your Opportunity: Internships and OJTs Workshop —** MSU-IIT | November 24, 2025.
- **Software Freedom Day 2023 —** MSU-IIT | September 16, 2023.